Module 2 -- Building Simple Circuits (FINAL PROJECT)

Lesson 16: Project -- The Lighthouse Circuit

★ Star-2-Star-3 Medium ⏰ 60--90 minutes (longer -- it is a project!)

👤 Age 6--12 years ⚡ **360 XP (the biggest XP haul of Module 2!)**

Your Mission Today

This is it, Circuit Explorer — the GRAND FINALE of Module 2! You are going to build a working **lighthouse** with a bright blinking LED, a hand-crafted tower, and a circuit that uses resistors, capacitors, and a transistor to flash automatically. You will use EVERYTHING you have learned: breadboarding, Ohm's Law, series and parallel knowledge, power selection, and — most importantly — your Magic Measurement Wand to test, verify, and debug every step. When you are done, you will have a real lighthouse you can put on your desk. Let us build something amazing!

Learning Objectives

This project brings together everything from Module 2:

- Breadboard layout and wiring (Lesson 9)
- Ohm's Law for resistor selection (Lesson 10)
- Series circuit concepts — components in a chain (Lesson 11)
- Understanding of parallel paths (Lesson 12)
- Voltage divider principles for timing (Lesson 13)
- ALL four multimeter modes for debugging (Lesson 14)
- Power source selection (Lesson 15)

What You Need

Item	Qty
9V battery + clip	1
Breadboard	1
Bright white or yellow LED	1
Red LED (for status indicator)	1
2N2222 NPN transistor	1
100-microfarad electrolytic capacitor	1
1000-microfarad electrolytic capacitor	1
Resistors: 330-ohm, 1k-ohm, 10k-ohm, 100k-ohm	1 each
Toggle switch	1
Jumper wires	10
Multimeter (Magic Measurement Wand)	1
Cardboard paper towel or toilet paper tube	1
White tissue paper or translucent plastic	1 piece
Tape (electrical + masking)	1 roll each
Scissors	1
Colored markers or paint (optional)	1 set
Small cardboard square (for the base)	1

What Are We Building?

A lighthouse that:

- Blinks a bright LED on and off automatically (no manual pressing!)
- Has an on/off switch
- Is built on a breadboard and housed in a paper tube tower
- Uses a capacitor + resistor + transistor to create the blinking effect

```
THE LIGHTHOUSE
==============

     (LED glows here)
    .---.
   /  O  \    <-- translucent paper diffuser
  |  LED  |
  |       |
  |  tube |   <-- paper towel tube
  |       |
  |       |
+---------+
| BREADBOARD | <-- circuit inside / on base
| + battery  |
+-----------+
```

How to Build It

Phase 1: Understand the Blinker Circuit (8 min)

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"The secret to blinking is a CAPACITOR. Remember from Module 1 how a capacitor charges up and then releases its energy? We are going to use that charge-discharge cycle to turn a transistor ON and OFF repeatedly — and that will blink the LED!"

How the blinker works (simplified):

```
The Blinker Cycle:

Step 1: Capacitor charges through a resistor (LED OFF)
         Voltage on capacitor rises slowly...

Step 2: When capacitor voltage reaches about 0.7V,
         the transistor switches ON

Step 3: Transistor ON = LED ON (lighthouse beam!)

Step 4: Capacitor discharges through the transistor
         Voltage drops below 0.7V

Step 5: Transistor switches OFF = LED OFF

Step 6: Capacitor starts charging again...
         (back to Step 1 -- the cycle repeats!)
```

> "The resistor and capacitor together control HOW FAST the blinking happens. Bigger resistor or bigger capacitor = SLOWER blink. Smaller = FASTER blink."

The time formula (for ages 9—12):

```
Blink time is roughly:  T = R x C

Example:
R = 100k ohm = 100,000 ohms
C = 100 microfarad = 0.0001 Farads

T = 100,000 x 0.0001 = 10 seconds per cycle

(In reality, the charge/discharge is not perfectly symmetric,
 so the ON and OFF times may differ. That is OK!)
```

Award: **+20 XP** **for understanding how the blinker circuit works!**

Phase 2: Build the Blinker Circuit (15 min)

Circuit diagram:

```
9V (+) ----[toggle switch]----+----[1k-ohm]----LED (+)----LED (-)----+
                              |                                      |
                              +----[100k-ohm]----+                   |
                              |                  |                   |
                              |              Base (B)                |
                              |                  |                   |
                              |         [2N2222 transistor]          |
                              |              |       |               |
                              |         Collector  Emitter           |
                              |              |       |               |
                              |              +       +---- GND -----+
                              |              |
                              +----[100uF cap (+)]----cap (-)---- GND
```

Simplified explanation:

> "The 100k-ohm resistor slowly charges the 100-microfarad capacitor. When the capacitor gets enough charge, the transistor turns on and the LED lights up. Then the capacitor releases its charge, the transistor turns off, and the LED goes dark. This repeats forever!"

Step-by-step breadboard build:

```
BREADBOARD LAYOUT:

Row 1-2: Power
- Toggle switch: one terminal in + rail, other terminal in row 2
- When switch is ON, row 2 gets 9V

Row 5: Timing resistor
- 100k-ohm resistor: row 2 to row 5

Row 5: Capacitor
- 100-microfarad capacitor: (+) leg in row 5, (-) leg in - rail
- WATCH POLARITY! Stripe (negative) goes to - rail

Row 5: Transistor Base
- 2N2222 transistor: Base (B) in row 5
- Collector (C) in row 8
- Emitter (E) in - rail (GND)

Row 8: LED circuit
- 1k-ohm resistor: row 2 to row 8 (connects to Collector)
- LED: Collector row -- actually, let us wire it clearly:
  - 1k-ohm resistor: one leg in row 2, other leg in row 10
  - LED: long leg (+) in row 10, short leg (-) in row 12
  - Jumper: row 12 to Collector row of transistor (row 8)

Battery:
- Red wire to + rail
- Black wire to - rail
```

"Take your time wiring this. Every connection matters. If one wire is in the wrong row, the whole circuit will not work."

Award: **+40 XP** **for building the complete blinker circuit!**

Phase 3: Wand Check — Verify Before Power On! (10 min)

"Before we flip the switch, let us make sure everything is connected correctly. Grab your Wand!"

Pre-power continuity checks (battery DISCONNECTED):

Set Wand to continuity mode.

```
| Test | From | To | Expected | Beep? |
|------|------|-----|----------|-------|
| 1 | Switch terminal 1 | + rail | Direct connection | YES |
| 2 | Switch terminal 2 | row 2 | Direct connection | YES |
| 3 | 100k resistor (row 2 end) | row 5 | Through resistor | YES |
| 4 | Capacitor (+) | row 5 | Direct connection | YES |
| 5 | Capacitor (-) | - rail | Direct connection | YES |
| 6 | Transistor Base | row 5 | Direct connection | YES |
| 7 | Transistor Emitter | - rail | Direct connection | YES |
| 8 | LED short leg row | Transistor Collector | Through wire | YES |
```

> "All beeps? Then the wiring is correct. No beep on any test? Check that connection — something is loose."

Component verification (still no power):

1. **Measure the 100k resistor:** Set Wand to ohms. Expected: about 100,000 ohms.
2. **Measure the 1k resistor:** Expected: about 1,000 ohms.
3. **Check LED polarity:** Long leg should be on the + side (closer to the resistor).
4. **Check capacitor polarity:** Stripe (negative) should face GND (- rail).
5. **Check transistor orientation:** Flat side facing you, legs should be E-B-C from left to right.

Award: **+40 XP** **for completing ALL pre-power verification checks!**

Phase 4: Power On and Test! (5 min)

> "Everything checked. Deep breath. Flip the switch!"

Connect the battery and flip the toggle switch ON.

What should happen:

The LED should start blinking! It may take a second or two for the first blink as the capacitor charges for the first time.

> 💬
> "IT BLINKS! Your lighthouse beam is working!"

Observe:

> 💬
> "How fast is it blinking? Count the blinks in 10 seconds."

If it does not blink (LED stays on or stays off), do not panic! We will debug in Phase 5.

Award: **+30 XP** **for a successfully blinking LED!**

Phase 5: Wand-Assisted Debugging (if needed) (10 min)

> 💬
> "If the LED does not blink, your Wand will find the problem. Here is your debugging workflow."

The Debugging Flowchart:

```
LED does not light at all:
|
+-- Measure battery: is it above 8V?
|   NO --> Replace battery
|   YES --> Continue
|
+-- Check switch: Wand continuity across switch (when ON)
|   NO BEEP --> Switch is off or broken
|   BEEP --> Continue
|
+-- Measure voltage at row 2 (after switch): should be about 9V
|   0V --> Switch wiring issue
|   9V --> Continue
|
+-- Check LED polarity: is long leg on + side?
|   NO --> Flip the LED
|   YES --> Continue
|
+-- Measure voltage across LED: should show 2-3V when lit
|   0V --> Transistor is not turning on; check base circuit
|   2-3V --> LED might be burned out; try a new one


LED stays ON (does not blink):
|
+-- Measure voltage at transistor Base (row 5) to GND
|   About 0.7V --> Transistor is permanently ON
|   Check capacitor polarity and 100k resistor value
|
+-- Is the capacitor connected correctly?
|   (+) to row 5, (-) to GND?
|
+-- Is the 100k resistor really 100k?
    Measure it! Maybe it is the wrong resistor.


LED stays OFF:
|
+-- Measure voltage at row 5 (base): does it rise over time?
|   YES but never reaches 0.7V --> 100k might be too large; try smaller
|   NO change --> Check capacitor and 100k resistor connections
```

Debugging journal:

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```
| Symptom | Wand Mode | Measurement | Problem Found | Fix Applied |
|---------|----------|------------|--------------|------------|
|         |          |            |              |            |
|         |          |            |              |            |
|         |          |            |              |            |
```

"Debugging is the MOST important skill in electronics. You just used EVERY Wand mode — voltage, continuity, resistance — to systematically find and fix a problem. This is EXACTLY what professional engineers do every day."

Award: **+40 XP** **for successfully debugging (or confirming everything works)!**

Phase 6: Experiment with Blink Speed (5 min)

"Now let us play with the blink speed!"

Experiment 1 — Faster blink:

Replace the 100-microfarad capacitor with a smaller value, or replace the 100k-ohm resistor with a smaller value (like 47k or 10k).

"What happened? It blinks faster! Smaller R or C = faster charging = faster blinking."

Experiment 2 — Slower blink:

Replace the 100-microfarad capacitor with the 1000-microfarad capacitor.

"Now it blinks much slower — like a real lighthouse! The bigger capacitor takes longer to charge."

Record your experiments:

```
| Resistor | Capacitor | Blink Speed (blinks per 10 sec) |
|----------|-----------|-------------------------------|
| 100k     | 100uF     |                               |
| 100k     | 1000uF    |                               |
| 10k      | 100uF     |                               |
```

Award: **+20 XP** **for experimenting with at least 2 different blink speeds!**

Phase 7: Build the Lighthouse Tower (10 min)

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"Now the fun part — building the lighthouse itself!"

Step 1: The tower

- Take the cardboard tube (paper towel roll)
- Optional: paint it with red and white stripes, or wrap with colored tape

Step 2: The light chamber

- At the top of the tube, tape a piece of white tissue paper or translucent plastic over the opening
- This is the "diffuser" — it spreads the LED light and makes it glow evenly

Step 3: Mount the LED

- Extend the LED wires (using jumper wires if needed) up through the tube
- Position the LED at the top, behind the tissue paper
- Tape or hot-glue the LED in place (adult helps with hot glue)

Step 4: The base

- Place the breadboard on a small piece of cardboard
- Stand the tube on the cardboard, next to the breadboard
- Route wires neatly from breadboard up into the tube

Step 5: Add the status LED (optional)

- Wire a small red LED (with 330-ohm resistor) in parallel from the + rail to - rail

- This stays ON whenever the switch is on, showing the lighthouse is "powered"

```
Finished Lighthouse:

     .----.
    / O O  \   <-- tissue paper diffuser
   |  (LED) |       with blinking LED behind it
   |--------|
   |  tube  |   <-- decorated tower
   |  tube  |
   |  tube  |
+--+--------+--+
| breadboard    |
| + battery     |
| + switch      |
| (red status LED) |
+--------------+
```

Award: **+30 XP** **for building the lighthouse tower!**

Phase 8: Final Wand Check — The Complete Diagnostic (5 min)

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"One last mission. Prove to me that EVERYTHING in your circuit is correct by taking these measurements."

Final diagnostic table:

```
| Measurement | Wand Mode | Expected | My Reading |
|------------|----------|---------|-----------|
| Battery voltage | DC Volts | 8.5--9.5V | |
| Voltage at row 2 (switch ON) | DC Volts | about 9V | |
| 100k resistor value | Ohms | about 100k | |
| 1k resistor value | Ohms | about 1k | |
| Voltage across LED (when ON) | DC Volts | 2--3.5V | |
| Base voltage (when LED is ON) | DC Volts | about 0.6--0.7V | |
| Continuity: switch ON | Continuity | BEEP | |
| Continuity: + rail to - rail | Continuity | NO BEEP | |
```

> 💬 "You just ran a COMPLETE diagnostic on your lighthouse circuit. You measured voltage, resistance, and continuity. You understand every part of this circuit. You are a true Circuit Builder!"

Award: **+30 XP** **for completing the full diagnostic!**

Module 2 Final Assessment

After building the lighthouse, have the kid answer these:

Circuit Knowledge:

1. "What is Ohm's Law?" (V = I x R)
2. "What is the difference between series and parallel circuits?" (Series = one path, parallel = multiple paths)
3. "In a voltage divider with equal resistors, what is Vout?" (Half of Vin)
4. "What makes the LED blink in the lighthouse?" (Capacitor charging and discharging through the transistor)

Wand Skills Check:

1. "Measure the battery." (DC Volts mode, probes in V/ohm and COM)
2. "Measure the 1k resistor." (Ohms mode, power off)
3. "Test this wire with continuity." (Continuity mode, listen for beep)
4. "Where would you put the probes to measure current through the LED?" (Break the circuit, insert Wand in series, red probe in mA jack)

Award: **+50 XP** **for completing the final assessment!**

Quick Quiz — Earn Bonus XP!

Question 1: Why does the LED in the lighthouse blink?

- A) The battery turns on and off
- B) A capacitor charges and discharges, turning a transistor on and off
- C) The switch has a timer inside

(Correct: B — +20 XP !)

Question 2: You want the lighthouse to blink SLOWER. What should you change?

- A) Use a SMALLER capacitor
- B) Use a BIGGER capacitor (or bigger resistor)
- C) Use a brighter LED

(Correct: B — bigger capacitor or resistor = slower charging = slower blink — +20 XP !)

Question 3: Your lighthouse LED does not light at all. What is the FIRST thing you should check with your Wand?

- A) Measure the LED voltage
- B) Measure the battery voltage to make sure it is charged
- C) Measure resistance of the resistor

(Correct: B — always check the power source first! — +20 XP !)

Module 2 Complete — Grand Finale!

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   MODULE 2 CHAMPION BADGE UNLOCKED!

   ____________________ has mastered
      Building Simple Circuits

   [check] Breadboard mastery
   [check] Ohm's Law (V = I x R)
   [check] Series circuits
   [check] Parallel circuits
   [check] Voltage dividers
   [check] ALL four multimeter modes
   [check] Power sources and batteries
   [check] Built a blinking lighthouse!

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Lesson 16 XP Breakdown:

Activity	XP
Understand blinker circuit	20
Build the blinker circuit	40
Pre-power Wand verification	40
Successful blink (power on)	30
Wand-assisted debugging	40
Blink speed experiments	20
Build the lighthouse tower	30
Final diagnostic	30
Final assessment	50
Quiz (3 questions)	60
TOTAL POSSIBLE	**360**

Module 2 Total XP Summary

Lesson	Title	Max XP
9	The Breadboard	230
10	Ohm's Law	240
11	Series Circuits	220
12	Parallel Circuits	230
13	Voltage Dividers	230
14	Using a Multimeter (Deep Dive)	260
15	Power Sources	210
16	Lighthouse Project	360
MODULE 2 TOTAL		**1,980**

Combined with Module 1 (2,030 XP): Grand Total Available = 4,010 XP

XP Ranks (cumulative):

- 0—500 XP: Circuit Apprentice
- 501—1,000 XP: Circuit Explorer
- 1,001—1,500 XP: Circuit Builder
- 1,501—2,000 XP: Circuit Engineer
- 2,001—2,500 XP: Circuit Scientist
- 2,501—3,500 XP: Circuit Wizard
- 3,501—4,010 XP: Circuit Master!

All Badges Earned in Module 2

Badge	Lesson
Breadboard Navigator	Lesson 9
Ohm's Law Apprentice	Lesson 10
Series Circuit Specialist	Lesson 11
Parallel Circuit Pro	Lesson 12
Voltage Divider Decoder	Lesson 13
Multimeter Master	Lesson 14
Power Source Expert	Lesson 15
Module 2 Champion	Lesson 16

Reflection Questions (Talk Through Together)

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"What was the coolest thing you built or discovered in Module 2?" "Which lesson was the hardest? What made it click?" "Which Wand mode are you MOST confident with now?" "Can you explain Ohm's Law to a friend?" "If you could change one thing about your lighthouse, what would it be?" "What do you think Module 3 will be about?"

What's Next — Module 3 Preview

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"You now know how to build real circuits, measure everything with your Wand, and debug like a pro! In Module 3, we enter the world of **integrated circuits and sensors**. You will meet the famous 555 timer chip, build a light-sensing circuit, and start connecting electronics to the real world. The journey from circuits to ROBOTS is getting closer!"

Parent/Instructor Notes

- This project takes longer than a regular lesson — plan for **60—90 minutes** or split across two sessions (build circuit in session 1, build tower + test in session 2).
- **The debugging phase (Phase 5) is the most valuable part of this lesson.** If the circuit works on the first try, consider intentionally introducing a bug (reversed LED, wrong resistor, loose wire) and having the child find it with the Wand.
- The blinker circuit described here is a simplified relaxation oscillator. It is not a perfect square wave, and blink timing may vary. That is fine — the learning goal is understanding RC timing and transistor switching, not precision.
- For a more reliable blinker, consider using a 555 timer IC (covered in Module 3). But the transistor + capacitor version is better for learning because every component has a clear role the child already understands.
- **Ages 6—8:** Focus on building the tower and watching it blink. The debugging and measurement phases can be done together with an adult guiding. The key win is "I built a lighthouse that blinks by itself!"
- **Ages 9—12:** Challenge them to predict the blink rate using T = R x C before building. Have them adjust values to hit a target blink rate (e.g., "Can you make it blink once every 5 seconds?").
- The lighthouse makes an excellent **show-and-tell** piece. Encourage the child to explain how the circuit works to family members.
- Take a photo or video of the finished lighthouse for their portfolio!
- **Celebrate completion of Module 2!** This is a major milestone. Consider a small reward, certificate, or special outing.